Description

Image





Caption

Isoprene is used for seals, tubing, vibration mounts, grips etc.

The material

Isoprene (or Polyisoprene) is synthetic natural rubber, and is processed in the same way as natural rubber. It has low hysteresis and high tear resistance, making it bouncy and tough.

Compositional summary

(CH2-C(CH3)-CH-CH2)n

General properties

Density		930	-	970	kg/m^3
Price	*	1.23	-	1.99	GBP/kg
Date first used		1960			

Mechanical properties

Young's modulus		0.0017	-	0.0026	GPa
Shear modulus		4e-4	-	6e-4	GPa
Bulk modulus	*	1.5	-	2	GPa
Poisson's ratio		0.499	-	0.5	
Yield strength (elastic limit)		23	-	25.8	MPa
Tensile strength		23	-	25.8	MPa
Compressive strength	*	27.6	-	31	MPa
Elongation		500	-	760	% strain
Fatigue strength at 10^7 cycles	*	9.2	-	10.3	MPa
Fracture toughness		0.133	-	0.204	MPa.m^0.5
Mechanical loss coefficient (tan delta)					

	0.07	-	0.13	

Thermal properties

Glass temperature	-75.2	-	-69.2	°C
Maximum service temperature	96.9	-	120	°C
Minimum service temperature	-60.2	-	-50.2	°C
Thermal conductor or insulator?	Good insulator			
Thermal conductivity	0.13	-	0.16	W/m.°C
Specific heat capacity	1.82e3	-	1.99e3	J/kg.°C
Thermal expansion coefficient	210	-	230	µstrain/°C

Electrical properties

Electrical conductor or insulator?	Good insulator			
Electrical resistivity	1e20	-	1e23	µohm.cm
Dielectric constant (relative permittivity)	2.3	-	3	
Dissipation factor (dielectric loss tangent)	1e-4	-	0.04	
Dielectric strength (dielectric breakdown)	16	-	23	MV/m

Optical properties

Transparency	Translucent			
Refractive index	1.51	-	1.53	

Critical Materials Risk

High critical material risk?	No			

Processability

Castability	4	-	5	
Moldability	4	-	5	
Machinability	3	-	4	
Weldability	1			

Durability: water and aqueous solutions

Water (fresh)	Excellent
Water (salt)	Excellent
Soils, acidic (peat)	Excellent
Soils, alkaline (clay)	Excellent
Wine	Acceptable

Durability: acids

Acetic acid (10%)	Acceptable

Acetic acid (glacial)	Limited use
Citric acid (10%)	Acceptable
Hydrochloric acid (10%)	Limited use
Hydrochloric acid (36%)	Limited use
Hydrofluoric acid (40%)	Limited use
Nitric acid (10%)	Limited use
Nitric acid (70%)	Unacceptable
Phosphoric acid (10%)	Acceptable
Phosphoric acid (85%)	Acceptable
Sulfuric acid (10%)	Acceptable
Sulfuric acid (70%)	Unacceptable

Durability: alkalis

Sodium hydroxide (10%)	Excellent
Sodium hydroxide (60%)	Limited use

Durability: fuels, oils and solvents

Amyl acetate	Unacceptable
Benzene	Unacceptable
Carbon tetrachloride	Unacceptable
Chloroform	Unacceptable
Crude oil	Unacceptable
Diesel oil	Unacceptable
Lubricating oil	Unacceptable
Paraffin oil (kerosene)	Unacceptable
Petrol (gasoline)	Unacceptable
Silicone fluids	Acceptable
Toluene	Unacceptable
Turpentine	Unacceptable
Vegetable oils (general)	Unacceptable
White spirit	Unacceptable

Durability: alcohols, aldehydes, ketones

Acetaldehyde	Limited use
Acetone	Acceptable
Ethyl alcohol (ethanol)	Acceptable
Ethylene glycol	Acceptable
Formaldehyde (40%)	Acceptable
Glycerol	Acceptable

Methyl alcohol (methanol)	Acceptable

Durability: halogens and gases

Chlorine gas (dry)	Unacceptable
Fluorine (gas)	Unacceptable
O2 (oxygen gas)	Unacceptable
Sulfur dioxide (gas)	Unacceptable

Durability: built environments

Industrial atmosphere	Excellent
Rural atmosphere	Excellent
Marine atmosphere	Excellent
UV radiation (sunlight)	Poor

Durability: flammability

Flammability	Highly flammable

Durability: thermal environments

Tolerance to cryogenic temperatures	Unacceptable
Tolerance up to 150 C (302 F)	Acceptable
Tolerance up to 250 C (482 F)	Unacceptable
Tolerance up to 450 C (842 F)	Unacceptable
Tolerance up to 850 C (1562 F)	Unacceptable
Tolerance above 850 C (1562 F)	Unacceptable

Primary material production: energy, climate change and water

Climate change (CO2-eq) (virgin grade production)	* 3.73	-	4.11	kg/kg
Embodied energy (virgin grade production)	* 81.8	-	90.2	MJ/kg
Water usage	* 139	-	153	l/kg

Material processing: energy

Polymer molding (injection, blow), embodied energy	* 16.1	-	17.7	MJ/kg
Machining, grinding, embodied energy	* 23.3	-	26.4	MJ/kg

Material processing: climate change

Polymer molding (injection, blow), climate change (CO2-eq)	* 1.05	-	1.16	kg/kg
Machining, grinding, climate change (CO2-eq)	* 1.52	-	1.73	kg/kg

Recycling and end of life

Recycle	✖

Functional recycle	✗			
Combust for energy recovery	✓			
Heat of combustion (net)	* 42.5	-	44.6	MJ/kg
Combustion CO2	* 3.15	-	3.31	kg/kg
Landfill	✓			
Biodegrade	✗			
Toxicity rating	Non-toxic			
A renewable resource?	✗			

Environmental notes

Isoprenes are thermosets, and thus cannot be recycled. Their disposal creates an environmental problem.

Recycle mark



Supporting information

Design guidelines

Isoprene has low hysteresis and high tear resistance, making it bouncy and tough.

Typical uses

Car tires and inner tubes, seals, belts, anti-vibration mounts, electrical insulation, tubing, rubber lining pipes and pumps, shoes.

Links

ProcessUniverse

Producers

Reference